Serial	FR ID	Functional Requirement	User Story ID	US Title	As a (precise role)	I want (requirement)	So that (completion condition)	User Story	Acceptance Criteria	Priority	Status
1	FR-01	The system shall support user registration and authentication	US-01.1	Create Account	Student	to create an account	I can access the platform and track my progress	As a Student, I want to create an account, so that I can access the platform and track my progress.	- Registration form includes fields for name, email, and password. - Registration can be done using Google, Github. - System validates unique email. - Password is stored securely (hashed). - Account is created with a default 'Student' role.	High	Done
2	FR-01	The system shall support user registration and authentication	US-01.2	Login to Platform	Student	to log in with my credentials	I can access my personalized dashboard	As a Student, I want to log in with my credentials, so that I can access my personalized dashboard.	- Login form accepts email and password. - System validates credentials against the database. - Upon success, user is redirected to the dashboard. - Upon failure, an appropriate error message is shown.	High	Done
3	FR-01	The system shall support user registration and authentication	US-01.3	Login to Platform	Admin	to log in with credentials	I can maintain platform security and user roles	As an Admin, I want to log in with credentials, so that I can maintain view and manage all registered users.	- Admin login has different interface. - Admin login is authenticated via pre-established email and password credentials. - After login the system shall redirect to admin dashboard.	High	Done
4	FR-02	The system shall allow students to discover and browse problems	US-02.1	Browse Problem List	Student	to view a list of all problems	I can find exercises to practice	As a Student, I want to view a list of all problems, so that I can find exercises to practice.	- A "Problems" page is accessible. - All problems with status 'Approved' are displayed. - Each problem shows its title, difficulty, and topic tags. - After clicking on the problems, the system redirects to that specific problem page	High	In Progress
5	FR-02	The system shall allow students to discover and browse problems	US-02.2	Filter and Search Problems	Student	to filter problems by difficulty and topic tags and search by title	I can find relevant challenges for my current topic	As a Student, I want to filter problems by difficulty and topic tags and search by title, so that I can find relevant challenges for my current topic.	- Filters for difficulty (Easy, Medium, Hard) and topic tags are available. - A search bar allows searching by problem title. - The list updates dynamically based on selected filters.	Medium	Done
6	FR-03	The system shall provide an in-browser code editor	US-03.1	Write Code in Editor	Student	to write and edit code directly in the browser	I can solve problems without leaving the page	As a Student, I want to write and edit codes directly in the browser, so that I can solve problems without leaving the page.	- Code editor is integrated into the problem page. - Editor has syntax highlighting and auto-indentation. - Student can write/type code directly into the editor.	Medium	In Progress
7	FR-04	The system shall provide code upload option and return verdicts	US-04.1	Submit Code for Evaluation	Student	to submit my code	I can get instant verdict on my solution	As a Student, I want to submit my code, so that I can get verdict on my solution.	- A "Submit" button is present on the problem page. - Code from editor is sent to backend via API. - System compiles and runs code against all hidden test cases. - System returns the verdict.	High	In Progress
8	FR-04	The system shall execute code and return instant verdicts	US-04.2	View Submission Verdict	Student	to see whether my solution was Accepted, Wrong Answer, or failed	I can know if my solution has any issuesl	As a Student, I want to see the status my solution was Accepted, Wrong Answer, or failed, so that I can know if my solution has any issuesl	- System returns one of the verdicts: AC, WA, TLE, RE, or CE. - Verdict is displayed prominently on the screen. - For any compilation related problem, error message is shown.	High	In Progress

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9	FR-05	The system shall provide a progress tracking dashboard	US-05.1	View Progress Dashboard	Student	to track my solved problems, success rate, and improvement over time	I can monitor my learning progress	As a Student, I want to track my solved problems, success rate, and improvement over time, so that I can monitor my learning progress.	- Personal dashboard is accessible from main menu. - A dedicated dashboard page is also available. - Displays total problems solved and overall success rate. - Shows breakdown of solved problems by difficulty. - Chart/graph shows progress.	Medium	In Progress
10	FR-06	The system shall store and display submission history	US-06.1	View Submission History	Student	to view a list of all my past submissions	I can review my past solutions and mistakes	As a Student, I want to view a list of all my past submissions, so that I can review my past solutions and mistakes.	- A "Submissions" page is accessible. - Displays list of all submissions. - Each entry shows problem name, timestamp, verdict, and code link. - Can click to view code for specific attempt.	Medium	In Progress
11	FR-07	The system shall allow students to bookmark problems	US-07.1	Bookmark Problems	Student	to bookmark interesting problems	I can practice them later	As a Student, I want to bookmark interesting problems, so that I can practice them later.	- A "Bookmark" button is present on each problem page. - Clicking toggles bookmark status for the problem. - Dedicated "Bookmarks" page lists all bookmarked problems.	Low	To Do
12	FR-08	The system shall display leaderboard via scoring of each users.	US-08.1	View Leaderboard	Student	to see my ranking compared to other students	I can foster healthy academic competition	As a Student, I want to see my ranking compared to other students, so that I can foster healthy academic competition.	- A "Leaderboard" page is accessible. - Lists all students ranked by problems solved and success rate. - Leaderboard updates in real-time.	Low	To Do
13	FR-09	The system shall allow problem setters to create and manage problems	US-09.1	Create New Problem	Problem Setter	to create a new coding problem	I can contribute content to the platform	As a Problem Setter, I want to create a new coding problem, so that I can contribute content to the platform.	- A "Create Problem" page is accessible with 'Problem Setter' role. - Form includes: Title, Description, Input/Output Format, Constraints, Difficulty, Topic Tags, Sample Test Cases, Video Link. - Problem is saved with 'Draft' status.	High	Done
14	FR-09	The system shall allow problem setters to create and manage problems	US-09.2	Manage My Problems	Problem Setter	to edit or delete problems I have created	I can maintain my content	As a Problem Setter, I want to edit or delete problems I have created, so that I can maintain my content.	- A "My Problems" page lists all problems I created. - Can click "Edit" to modify any detail. - Can click "Delete" to remove problem permanently.	Medium	In Progress
15	FR-10	The system shall allow problem submission and review workflow	US-10.1	Submit Problem for Review	Problem Setter	to submit my problem to the admin for approval	It can be published on the platform for students	As a Problem Setter, I want to submit my problem to the admin for approval, so that it can be published on the platform for students.	- A "Submit for Review" button is available on problem page. - Clicking changes status to 'Pending'. - Problem is sent to Admin's moderation queue.	High	In Progress
16	FR-10	The system shall allow problem submission and review workflow	US-10.2	Review Pending Problems	Admin	to check the list of all submitted problems	It can be reviewd for approval/rejection.	As an admin, I want to check the list of all submitted problems, so that I can review them for approval/rejection.	- A "Pending Problems" section is accessible to the Admin. - Displays a list of all problems with 'Pending' status. - Each entry shows problem title, setter name, and problem description. - Admin can select a problem to view its full details for review.	High	In Progress
17	FR-10	The system shall allow problem submission and review workflow	US-10.3	Track Problem Status	Problem Setter	to track whether my problem is Draft, Pending, Approved, or Rejected	I can know if action is needed	As a Problem Setter, I want to track whether my problem is Draft, Pending, Approved, or Rejected, so that I can know if action is needed.	- "My Problems" page clearly shows status of each problem. - If rejected, feedback from admin is visible.	High	In Progress

Serial	FR ID	Functional Requirement	User Story ID	US Title	As a (precise role)	I want (requirement)	So that (completion condition)	User Story	Acceptance Criteria	Priority	Status
18	FR-11	The system shall allow admin to moderate problems	US-11.1	Review Pending Problems	Admin	to review all problems submitted by problem setters	I can ensure quality and correctness of content	As an Admin, I want to review all problems submitted by problem setters, so that I can ensure quality and correctness of content.	- Admin Dashboard has section for pending problems. - Can click to view full details of pending problem.	High	In Progress
19	FR-11	The system shall allow admin to moderate problems	US-11.2	Approve Problems	Admin	to approve problems	They become visible to students on the platform	As an Admin, I want to approve problems, so that they become visible to students on the platform.	- An "Approve" button is present on problem review page. - Clicking changes problem status to 'Approved'. - Problem appears in Student problem list.	High	In Progress
20	FR-11	The system shall allow admin to moderate problems	US-11.3	Reject Problems with Feedback	Admin	to reject problems and provide feedback to the setter	The setter can correct and resubmit	As an Admin, I want to reject problems and provide feedback to the setter, so that the setter can correct and resubmit.	- A "Reject" button is present on problem review page. - Admin can type a feedback message. - Submitting changes status to 'Rejected' and sends feedback.	High	In Progress
21	FR-12	The system shall support problem tagging.	US-12.1	Add Topic Tags to Problems	Problem Setter	to add topic tags (e.g., Arrays, Recursion, Strings) when creating a problem	students can easily find problems by topic	As a Problem Setter, I want to add topic tags (e.g., Arrays, Recursion, Strings) when creating a problem, so that students can easily find problems by topic.	- Predefined list of topic tags is available. - Setter can select multiple tags per problem. - Tags appear on problem listing page. - Students can filter by tags.	Medium	Done
22	FR-13	The system shall support problem difficulty ratings	US-13.1	Set Problem Difficulty	Problem Setter	to assign a difficulty level (Easy, Medium, Hard) to my problem	students can choose problems matching their skill level	As a Problem Setter, I want to assign a difficulty level (Easy, Medium, Hard) to my problem, so that students can choose problems matching their skill level.	- Dropdown menu with Easy, Medium, Hard options. - Difficulty is displayed on problem listing. - Students can filter by difficulty. - Difficulty can be edited later.	Medium	Done
23	FR-14	The system shall provide submission analytics to problem setters	US-14.1	View Problem Analytics	Problem Setter	to see how many students attempted and solved my problems	I can understand problem difficulty and quality	As a Problem Setter, I want to see how many students attempted and solved my problems, so that I can understand problem difficulty and quality.	- An "Analytics" page is accessible. - Shows number of students who attempted the problem. - Shows breakdown of verdicts (AC, WA, TLE, etc.). - Highlights most common errors.	Low	To Do
24	FR-15	The system shall allow admin to manage contests	US-15.1	Create Contest	Admin	to create a coding contest with selected problems and time limits	I can organize competitive events for students	As an Admin, I want to create a coding contest with selected problems and time limits, so that I can organize competitive events for students.	- A "Create Contest" page is accessible. - Can select a set of approved problems. - Can set start time, end time, and duration. - Contest is published to students after creation.	Low	Done
25	FR-15	The system shall allow admin to manage contests	US-15.2	View Contest Leaderboard	Student	to see real-time rankings during a contest	I can know my performance and rank	As a Student, I want to see real-time rankings during a contest, so that I can know my performance and rank.	- Contest page has a live leaderboard. - Ranking based on score and penalty time (ICPC model). - Leaderboard updates automatically.	Low	To Do
26	FR-15	The system shall allow admin to manage contests	US-15.3	Participate in Contest	Student	to participate in a contest	I can compete with others	As a Student, I want to participate in a contest, so that I can compete with others.	- A 'Participate' button is available on the contest page. - User can submit solutions for the problems in the contest. - Submissions are evaluated and scores are updated on the live leaderboard. - User can view their current rank and score during the contest.	Low	To Do

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27	FR-16	The system shall support notifications for important events	US-16.1	Receive Problem Approval Notification	Problem Setter	to receive an notification when my problem is approved or rejected	I know the status of my submission.	As a Problem Setter, I want to receive an notification when my problem is approved or rejected, so that I know the status of my submission.	- Notification is sent when problem status changes to 'Approved' or 'Rejected'. - Notification includes problem title, status, and admin feedback (if rejected). - Notification is sent to the setter's registered email address. - Notification template is professional and includes platform branding.	Medium	To Do
28	FR-16	The system shall support notifications for important events	US-16.2	Receive Status Updates	Admin	to receive notifications on problem submissions and role requests	I can review them promptly	As an Admin, I want to receive notifications on problem submissions and role requests, so that I can review them promptly.	- Notification is sent when a new problem is submitted. - Notification is sent when a role change is requested. - Notifications appear in the admin dashboard.	Medium	To Do
29	FR-16	The system shall support notifications for important events	US-16.3	Receive Platform Notifications	Student	to receive notifications regarding contests and role changes	I can stay updated on platform activities	As a Student, I want to receive notifications regarding contests and role changes, so that I can stay updated on platform activities.	- Notification is sent when a new contest is created. - Notification is sent when a role change request is approved or rejected. - Notifications appear in the student dashboard.	Medium	To Do
30	FR-17	The system shall support role change requests from users	US-17.1	Request Problem Setter Role	Student	to request to become a Problem Setter	I can create and contribute problems to the platform	As a Student, I want to request to become a Problem Setter, so that I can create and contribute problems to the platform.	- A "Request Role Change" button is available in user profile. - Request is sent to Admin for approval. - User can see pending request status. - Notification is sent to Admin about new request.	High	Done
31	FR-17	The system shall support role change requests from users	US-17.2	Approve/Reject Role Requests	Admin	to approve or reject role change requests from students	I can control who becomes a Problem Setter	As an Admin, I want to approve or reject role change requests from students, so that I can control who becomes a Problem Setter.	- Admin dashboard has a section for pending role requests. - Admin can click "Approve" or "Reject" for each request. - User receives notification about decision. - If approved, user's role changes to 'Problem Setter'.	High	In Progress
32	FR-18	The system shall allow students to view problem details and sample test cases	US-18.1	View Problem Description	Student	to view detailed problem description, input/output format, and sample test cases	I understand what the problem expects and can solve it correctly	As a Student, I want to view detailed problem description, input/output format, and sample test cases, so that I understand what the problem expects and can solve it correctly.	- Problem page displays title, description, input/output format, and constraints. - Sample test cases with expected outputs are shown. - Topic tags and difficulty level are displayed. - Problem statement is well-formatted with code blocks and examples.	High	In Progress
33	FR-19	The system shall provide a user profile page	US-19.1	View Profile	Student	to view my profile page with personal information	I can see my basic info & achievements.	As a Student, I want to view my profile page with personal information, so that I can see my basic info & achievements.	- Profile page shows name, email, and join date. - Shows total problems solved and success rate. - Shows recent submissions. - Shows badges or achievements (optional).	Medium	Done
34	FR-19	The system shall provide a user profile page	US-19.2	Update Profile	Student	to update my profile information	my personal details are up to date	As a Student, I want to update my profile information, so that my personal details are up to date.	- Profile page has an "Edit" button. - Fields like name and password can be changed. - Email changes require verification. - Changes are saved to database.	Low	To Do

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35	FR-20	The system shall support code compilation error messages	US-20.1	View Compilation Errors	Student	to see detailed compilation error messages when my code fails to compile	I can fix the errors and resubmit	As a Student, I want to see detailed compilation error messages when my code fails to compile, so that I can fix the errors and resubmit.	- Compilation errors are displayed clearly. - Error message includes line number and description. - Error messages are formatted for readability. - Common errors (e.g., missing semicolon) are highlighted.	High	In Progress
37	FR-21	The system shall support reviewing and approving problem difficulty ratings.	US-21.1	Review and Assign Problem Difficulty	Admin	to review proposed difficulty ratings and formally assign them to problems	the platform maintains a consistent, accurate, and balanced difficulty standard across all challenges	As an Admin, I want to review the difficulty levels suggested by problem setters and formally assign or adjust them, so that the platform maintains a consistent and accurate difficulty standard across all challenges.	- Dashboard/queue displaying pending or community-rated problem difficulties. - Ability for Admin to overwrite or confirm the proposed rating (Easy, Medium, Hard). - Once approved, the updated difficulty instantly reflects across the problem listings for students.	Medium	To Do
38	FR-22	The system shall support handling and displaying Time Limit Exceeded (TLE) execution errors.	US-22.1	View Timeout Errors	Student	to see exactly when my code exceeds the allocated execution time limit	I can optimize my code's time complexity and algorithms before resubmitting.	As a Student, I want to see a clear Time Limit Exceeded (TLE) status when my code runs longer than allowed, so that I can optimize my code's time complexity and algorithms before resubmitting.	- The system stops execution immediately when the per-testcase time limit is breached. - The status is clearly displayed as "Time Limit Exceeded" or "TLE".	High	In Progress